usb_power
File: usb_power.kicad_sch
USB-C sink (no host), VBUS protection, USB0 device/FEL

power
File: power.kicad_sch
Regulators, rail order, supervisors, staged-power links

f133
File: f133.kicad_sch
F133-B SoC: supplies, clock, reset, pin assignment

boot_nor
File: boot_nor.kicad_sch
32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery

mn820
File: mn820.kicad_sch
MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel

fpga
File: fpga_ecp5.kicad_sch
LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation

psram
File: psram.kicad_sch
APS6404L 8 MiB PSRAM: switched VDD (DONE-gated), static pulls

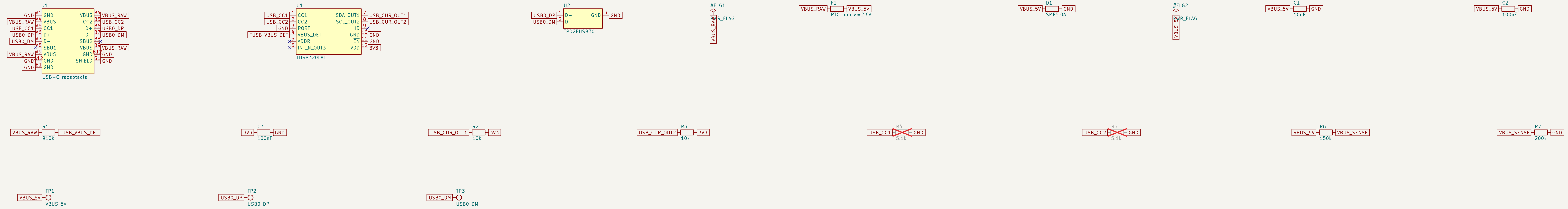
sd_audio
File: sd_audio.kicad_sch
microSD game media (not boot), 2x MAX98357A I2S speakers

controls
File: controls.kicad_sch
Game buttons, LEDs, debug UART

NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R85.kicad_sch
Title: ARC-1 Compact R8.5 – hierarchical root
Size: A3
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
Sheet generated from design.py; global labels carry every net.



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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1

Sheet: /usb_power/
File: usb_power.kicad_sch

Title: ARC-1 Compact R8.5: USB-C sink (no host), VBUS protection, USB0 d

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 2/10

Regulators, rail order, supervisors, staged−power links
Sheet generated from design.py; global labels carry every net.

U3 3V3 2A: VBUS_5V 1 EN, GND 2 GND, SW_3V3 3 SW, VIN 4 VBUS_5V, PG 5 PG_3V3_BUCK, FB 6 FB_3V3
U4 TPS3808G33: PG_3V3 1 RESET, GND 2 GND, SUP33_MR_N 3 MR, CT 4 SUP33_CT, SENSE 5 3V3, VDD 6 3V3
U5 VDD_CORE 0.95V: PG_3V3 1 EN, GND 2 GND, SW_CORE 3 SW, VIN 4 VBUS_5V, FB 5 FB_CORE
U6 1V8_DRAM: PG_3V3 1 EN, GND 2 GND, SW_DRAM 3 SW, VIN 4 VBUS_5V, FB 5 FB_DRAM
U7 1V8_AN: 3V3 1 IN, GND 2 GND, 3V3 3 EN, NC 4, OUT 5 1V8_AN
U8 TPS3808G09: F133_RESET_N 1 RESET, GND 2 GND, RST_BTN_N 3 MR, CT 4 SUPC_CT, SENSE 5 VDD_CORE, VDD 6 3V3
U9 1V1_FPGA: FPGA_PWR_EN 1 EN, GND 2 GND, SW_1V1 3 SW, VIN 4 VBUS_5V, PG 5 FPGA_VCC_PG, FB 6 FB_1V1
U10 3V3_FPGA switch: 3V3 1 VIN, GND 2 GND, FPGA_PWR_EN 3 ON, CT 4 SW_FPGA_CT, QOD 5 3V3_FPGA, VOUT 6 3V3_FPGA
U30 2V5_FPGA_AUX: 3V3_FPGA 1 IN, GND 2 GND, 3V3_FPGA 3 EN, NC 4, OUT 5 2V5_AUX_REG
U11 3V3_MN switch: 3V3 1 VIN, GND 2 GND, MN_PWR_EN 3 ON, CT 4 SW_MN_CT, QOD 5 3V3_MN, VOUT 6 3V3_MN
U12 PSRAM_VDD switch: 3V3_FPGA 1 VIN, GND 2 GND, PSRAM_EN 3 ON, CT 4 SW_PS_CT, QOD 5 PSRAM_VDD, VOUT 6 PSRAM_VDD
U13 AND: PSRAM_PWR_REQ 1 A, DONE_DLY 2 B, GND 3 GND, Y 4 PSRAM_EN, VCC 5 3V3
U14 TPS3808G30: PSRAM_PG 1 RESET, GND 2 GND, PSRAM_SUP_MR_N 3 MR, CT 4 PSRAM_SUP_CT, SENSE 5 PSRAM_VDD, VDD 6 3V3_FPGA
J11 FPGA power override: 3V3 1, FPGA_PWR_EN 2, 3 GND
SW1 RESET: RST_BTN_N 1, 2 GND
C4 VBUS_5V 10uF GND
L1 SW_3V3 2.2uH 3V3_REG
R8 3V3_REG 453k FB_3V3
R9 FB_3V3 100k GND
C5 3V3_REG 22uF GND
C6 3V3_REG 22uF GND
R10 PG_3V3_BUCK 100k 3V3_REG
#FLG3 PWR_FLAG 3V3_REG
R11 3V3_REG 0R 3V3
#FLG4 PWR_FLAG 3V3
TP4 3V3_REG
TP5 3V3
TP6 PG_3V3_BUCK
C7 SUP33_CT 2.2nF GND
R12 SUP33_MR_N 100k 3V3
R13 PG_3V3 10k 3V3
C8 3V3 100nF GND
C9 VBUS_5V 10uF GND
L2 SW_CORE 2.2uH VDD_CORE_REG
R14 VDD_CORE_REG 57.6k FB_CORE
R15 FB_CORE 100k GND
C10 VDD_CORE_REG 22uF GND
C11 VDD_CORE_REG 22uF GND
#FLG5 PWR_FLAG VDD_CORE_REG
R16 VDD_CORE_REG 0R VDD_CORE
#FLG6 PWR_FLAG VDD_CORE
TP7 VDD_CORE
C12 VBUS_5V 10uF GND
L3 SW_DRAM 2.2uH 1V8_DRAM
R17 1V8_DRAM 200k FB_DRAM
R18 FB_DRAM 100k GND
C13 1V8_DRAM 22uF GND
#FLG7 PWR_FLAG 1V8_DRAM
TP8 1V8_DRAM
C14 3V3 1uF GND
C15 1V8_AN 1uF GND
TP9 1V8_AN
C16 SUPC_CT 15nF GND
R19 RST_BTN_N 100k 3V3
R20 F133_RESET_N 10k 1V8_LDOA
TP10 F133_RESET_N
TP11 PG_3V3
C17 VBUS_5V 10uF GND
L4 SW_1V1 2.2uH 1V1_FPGA_REG
R21 1V1_FPGA_REG 82.5k FB_1V1
R22 FB_1V1 100k GND
C18 1V1_FPGA_REG 22uF GND
C19 1V1_FPGA_REG 22uF GND
#FLG8 PWR_FLAG 1V1_FPGA_REG
R23 1V1_FPGA_REG 0R 1V1_FPGA
#FLG9 PWR_FLAG 1V1_FPGA
R24 FPGA_PWR_EN_F 1k FPGA_PWR_EN
R25 FPGA_PWR_EN 100k GND
R26 FPGA_VCC_PG 100k 3V3
TP12 1V1_FPGA
TP13 FPGA_VCC_PG
TP14 FPGA_PWR_EN
C20 SW_FPGA_CT 1nF GND
C21 3V3 1uF GND
C22 3V3_FPGA 10uF GND
TP15 3V3_FPGA
C23 3V3_FPGA 1uF GND
C24 2V5_AUX_REG 1uF GND
FB2 2V5_AUX_REG ferrite 120R@100MHz 2V5_FPGA_AUX
#FLG10 PWR_FLAG 2V5_FPGA_AUX
TP16 2V5_FPGA_AUX
C25 SW_MN_CT 1nF GND
C26 3V3 1uF GND
C27 3V3_MN 22uF GND
R27 MN_PWR_EN 100k 3V3
TP17 3V3_MN
C28 SW_PS_CT 1nF GND
C29 3V3_FPGA 1uF GND
C30 PSRAM_VDD 4.7uF GND
C31 3V3 100nF GND
R28 PSRAM_PWR_REQ 100k GND
R29 PSRAM_EN 100k GND
C32 PSRAM_SUP_CT 1nF GND
R30 PSRAM_SUP_MR_N 100k 3V3_FPGA
R31 PSRAM_PG 10k 3V3_FPGA
C33 3V3_FPGA 100nF GND
TP18 PSRAM_VDD
TP19 PSRAM_PG
TP20 PSRAM_EN
TP21 GND
TP22 GND
TP23 GND
TP24 GND
#FLG11 PWR_FLAG GND
FID1 fiducial
FID2 fiducial
FID3 fiducial
H1 GND M2.5
H2 GND M2.5
H3 GND M2.5
H4 GND M2.5

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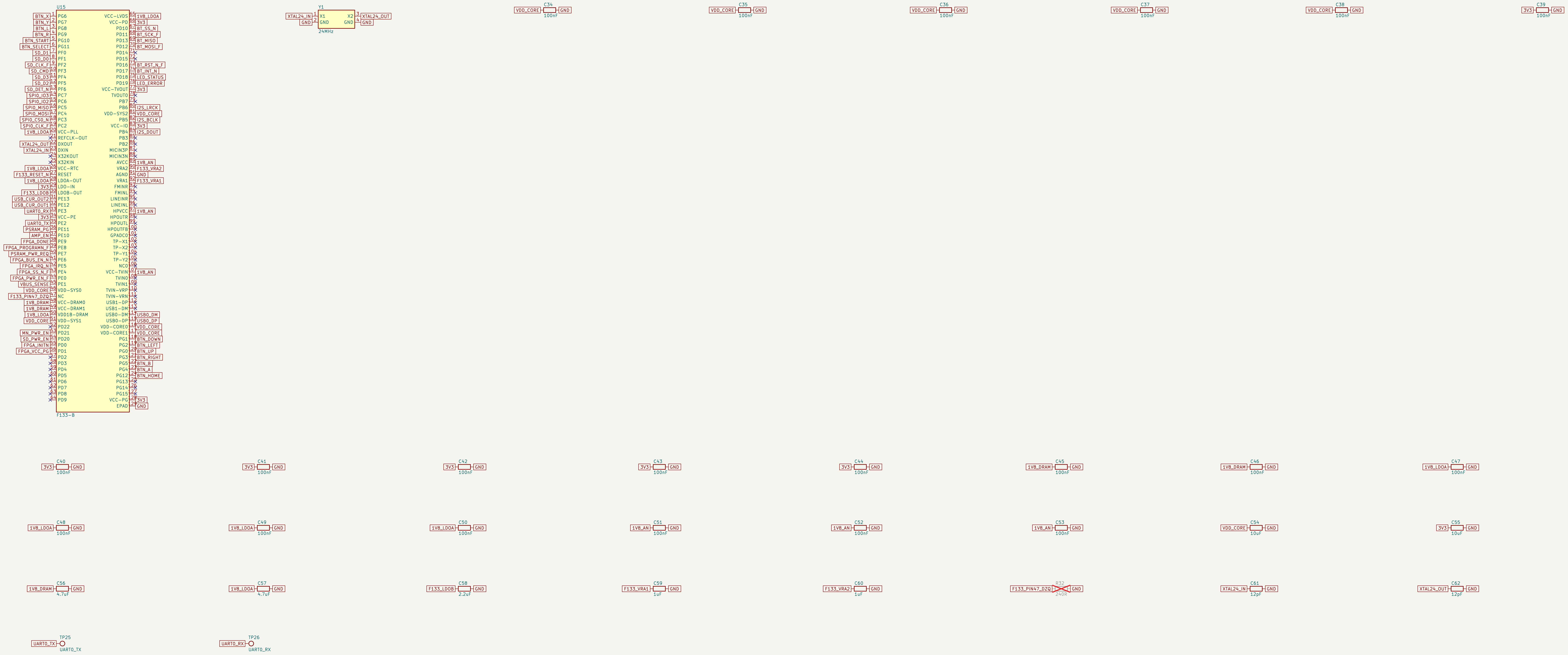
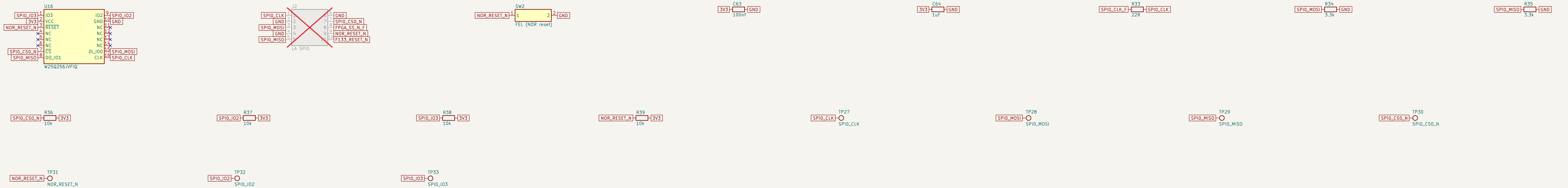
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /power/
File: power.kicad_sch
Title: ARC−1 Compact R8.5: Regulators, rail order, supervisors, staged−powe
Size: User | Date: 2026−10−15 | **Rev: R85−draft**
KiCad E.D.A. 9.0.9 | Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /f133/
File: f133.kicad_sch
Title: ARC-1 Compact R8.5: F133-B SoC: supplies, clock, reset, pin assignm
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 4/10

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.



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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

STRUTHIO ARC-1

Sheet: /boot_nor/
File: boot_nor.kicad_sch

Title: ARC-1 Compact R8.5: 32 MiB SPI NOR boot (BOOT-SEL 00), straps, FI

Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

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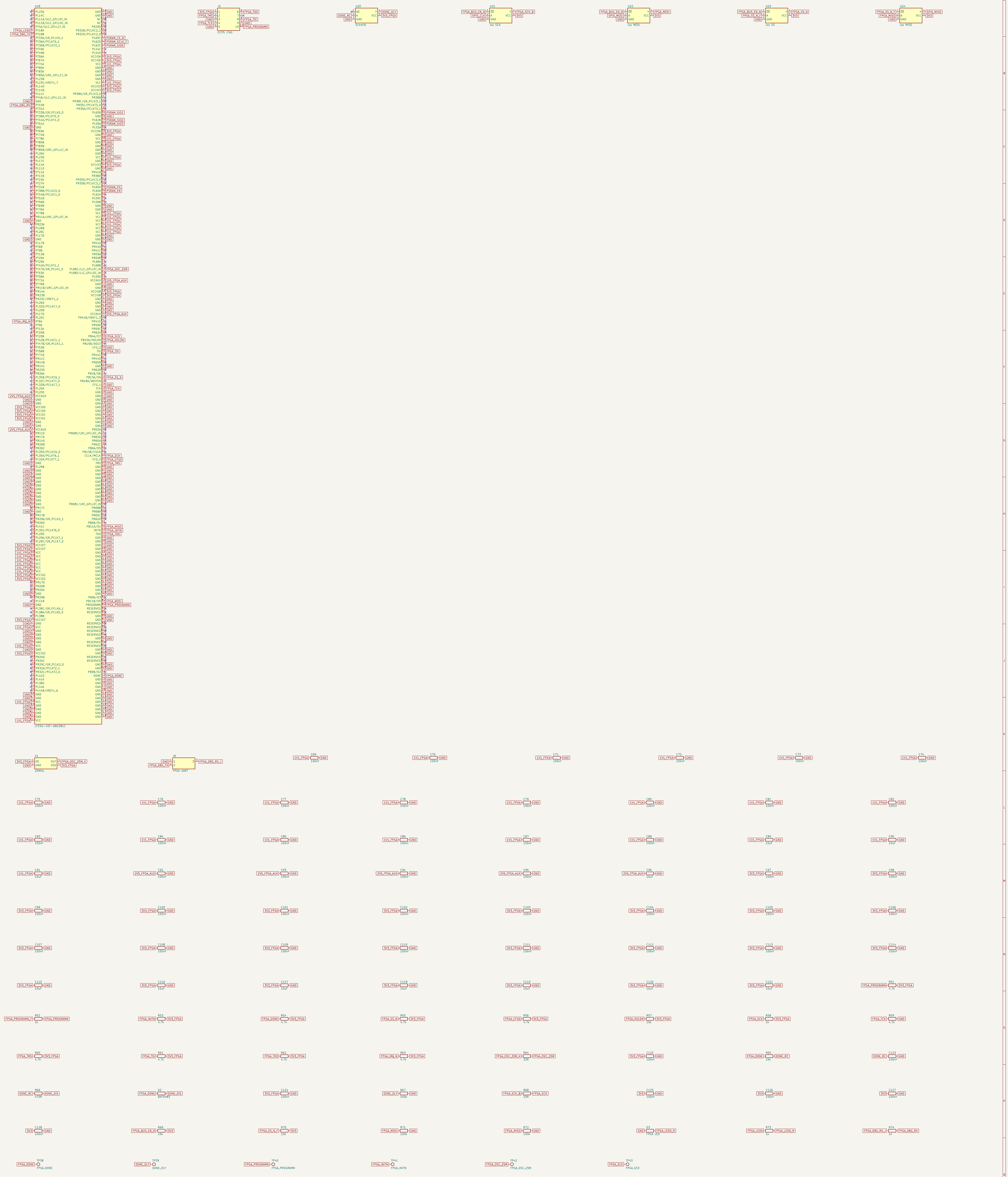
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /mn820/
File: mn820.kicad_sch
Title: ARC-1 Compact R8.5: MN820 (BT820B + 128 MiB DDR3L) module sock
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 6/10

LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation
Sheet generated from design.py; global labels carry every net.



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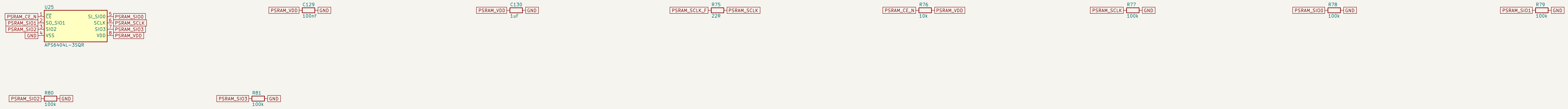
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /fpga/
File: fpga_ecp5.kicad_sch

Title: ARC-1 Compact R8.5: LFE5U-45F-6BG381C (optional accelerator): rai

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD (DONE-gated), static pulls
Sheet generated from design.py; global labels carry every net.



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SELECTED_UNVERIFIED engineering capture - NOT released for fabrication

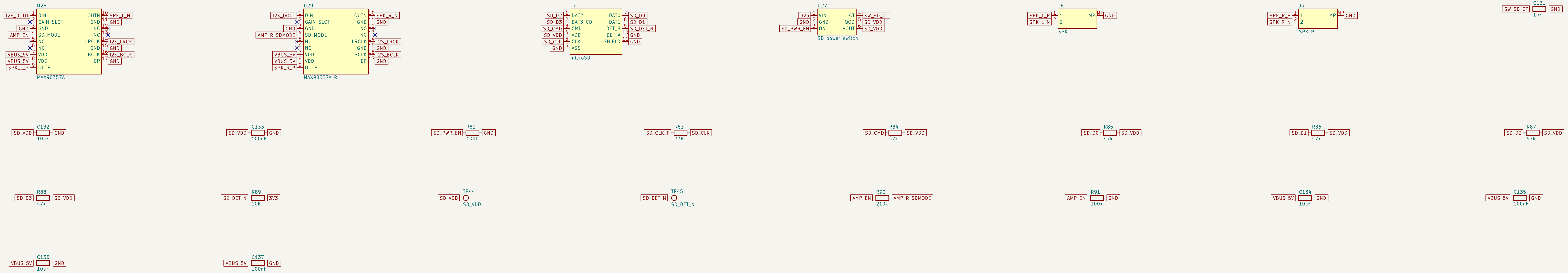
STRUTHIO ARC-1

Sheet: /psram/
File: psram.kicad_sch

Title: ARC-1 Compact R8.5: APS6404L 8 MiB PSRAM: switched VDD (DONE-g

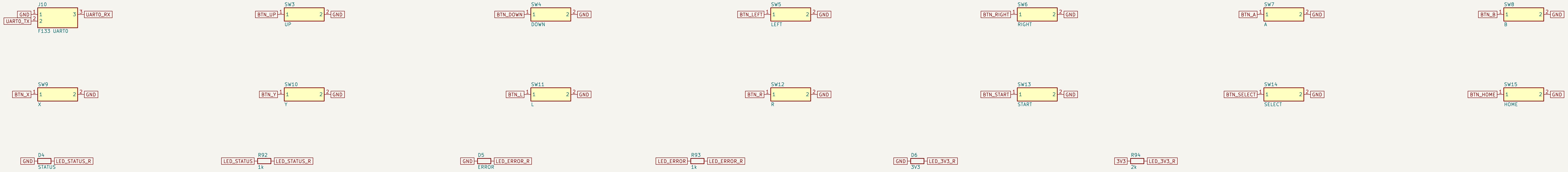
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.



Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R8.5: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py; global labels carry every net.



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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /controls/
File: controls.kicad_sch
Title: ARC-1 Compact R8.5: Game buttons, LEDs, debug UART
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 10/10